

ALL τ_k -MAXIMAL GRAPHS HAVE THE SAME SIZE

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ABSTRACT. A graph is τ_k -maximal if none of its subgraphs has $k + 1$ edge-disjoint spanning trees, but adding any missing edge creates such a subgraph. Wang and Tian conjectured that every τ_k -maximal graph on $n \geq 2k + 2$ vertices has exactly $(k + 1)(n - 1) - 1$ edges. We prove this conjecture.

1. INTRODUCTION

All graphs in this note are finite and simple. For a graph H with at least two vertices, $\tau(H)$ denotes the largest number of edge-disjoint spanning trees of H , and for a graph G we put

$$\bar{\tau}(G) = \max\{\tau(H) : H \subseteq G, |V(H)| \geq 2\}.$$

Adding an edge raises $\bar{\tau}$ by at most one. Following Wang and Tian [6], a graph G is τ_k -maximal if $\bar{\tau}(G) \leq k$ while $\bar{\tau}(G + e) \geq k + 1$ for every edge e of the complement of G . Since $\tau(K_n) = \lfloor n/2 \rfloor$, the complete graph K_n is the only τ_k -maximal graph on $n \leq 2k + 1$ vertices, so the interesting range is $n \geq 2k + 2$. There Wang and Tian proved that a τ_k -maximal graph has at most $(k + 1)(n - 1) - 1$ edges, built a family of τ_k -maximal graphs with exactly that many edges, and proved that every τ_1 -maximal graph on n vertices has $2n - 3$ edges. They proposed the following statement [6, Conjecture 1].

Conjecture 1. Let $k \geq 1$ and $n \geq 2k + 2$. Every τ_k -maximal graph of order n has exactly $(k + 1)(n - 1) - 1$ edges.

We prove it, together with a description of the τ_k -maximal graphs by counting alone. For $X \subseteq V(G)$ let $e_G(X)$ be the number of edges of G with both ends in X .

Theorem 1.1. Let $k \geq 1$ and $n \geq 2k + 2$, and let G be a graph with n vertices. The following are equivalent.

- (a) G is τ_k -maximal.
- (b) $|E(G)| = (k + 1)(n - 1) - 1$, and $e_G(X) \leq (k + 1)(|X| - 1) - 1$ for every $X \subseteq V(G)$ with $|X| \geq 2$.

In particular, Conjecture 1 is true.

The proof has three steps. Section 2 turns the condition $\bar{\tau}(G) \leq k$ into a counting condition, using the theorem of Nash-Williams and Tutte. Section 3 shows, by an exchange argument with tight sets, that all maximal edge sets satisfying the counting condition have the same size; in the language of matroids they are the bases of a count matroid. Section 4 computes the common size with one explicit example.

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2. THE COUNTING CONDITION

Throughout, $a = k + 1 \geq 2$. Call a graph G a -sparse if

$$(2.1) \quad e_G(X) \leq a(|X| - 1) - 1 \quad \text{for every } X \subseteq V(G) \text{ with } |X| \geq 2.$$

We use the theorem of Nash-Williams [4] and Tutte [5]: a graph H has a edge-disjoint spanning trees if and only if, for every partition $\mathcal{P}$ of $V(H)$ into t parts, at least $a(t - 1)$ edges of H join different parts.

Lemma 2.1. *For every graph G , $\bar{\tau}(G) \leq k$ if and only if G is $(k + 1)$ -sparse.*

Proof. If a subgraph H of G on a vertex set X with $|X| \geq 2$ has a edge-disjoint spanning trees, then $e_G(X) \geq |E(H)| \geq a(|X| - 1)$, and (2.1) fails for X .

Conversely, suppose (2.1) fails, and choose a set X with $|X| \geq 2$ and $e_G(X) \geq a(|X| - 1)$ with $|X|$ as small as possible. Let $\mathcal{P} = \{P_1, \dots, P_t\}$ be a partition of X with $t \geq 2$. A part with at least two vertices is a smaller set than X , so by the choice of X it spans at most $a(|P_i| - 1) - 1 < a(|P_i| - 1)$ edges; a part with one vertex spans $0 = a(|P_i| - 1)$ edges. Hence the number of edges of $G[X]$ joining different parts is at least

$$e_G(X) - \sum_{i=1}^t a(|P_i| - 1) \geq a(|X| - 1) - a(|X| - t) = a(t - 1).$$

By the theorem of Nash-Williams and Tutte, $G[X]$ has a edge-disjoint spanning trees, so $\bar{\tau}(G) \geq a = k + 1$. $\square$

By Lemma 2.1, a graph G on the vertex set V of K_n is τ_k -maximal exactly when $E(G)$ is $(k + 1)$ -sparse and $E(G) \cup \{e\}$ is not, for every edge e of K_n outside $E(G)$. In other words, $E(G)$ is a maximal sparse subset of $E(K_n)$; here and below we call a set F of edges sparse when the graph (V, F) is a -sparse.

3. TIGHT SETS

Fix a sparse set I of edges of K_n , and write $e(X) = e_I(X)$. A set $X \subseteq V$ with $|X| \geq 2$ is *tight* if $e(X) = a(|X| - 1) - 1$, that is, if (2.1) holds for X with equality.

Lemma 3.1. (i) $e(X \cup Y) + e(X \cap Y) \geq e(X) + e(Y)$ for all $X, Y \subseteq V$.

(ii) If X and Y are tight and $|X \cap Y| \geq 2$, then $X \cup Y$ is tight.

(iii) If uv is an edge of K_n outside I and $I \cup \{uv\}$ is not sparse, then some tight set contains both u and v .

Proof. (i) An edge counted in $e(X)$ or in $e(Y)$ lies inside $X \cup Y$, and an edge counted in both lies inside $X \cap Y$.

(ii) Since $|X \cap Y| \geq 2$, the set $X \cap Y$ satisfies (2.1), and by (i)

$$e(X \cup Y) \geq (a(|X| - 1) - 1) + (a(|Y| - 1) - 1) - (a(|X \cap Y| - 1) - 1) = a(|X \cup Y| - 1) - 1.$$

The reverse inequality is (2.1) for $X \cup Y$.

(iii) Some X with $|X| \geq 2$ has more than $a(|X| - 1) - 1$ edges of $I \cup \{uv\}$ inside it, and at most $a(|X| - 1) - 1$ edges of I . So uv lies inside X , and X is tight. $\square$

Lemma 3.2. *Let I be a maximal sparse subset of $E(K_n)$ and let J be any sparse subset. Then $|J| \leq |I|$.*

Proof. Tight sets refer to I . Let $M_1, \dots, M_p$ be the tight sets that are maximal under inclusion; every tight set lies inside one of them. If $|M_i \cap M_j| \geq 2$, then $M_i \cup M_j$ is tight and contains both, so $M_i = M_i \cup M_j = M_j$. Hence two different sets M_i share at most one vertex, and every edge of K_n lies inside at most one of them. Moreover, if $uv \notin I$, then $I \cup \{uv\}$ is not sparse because I is maximal, so by Lemma 3.1(iii) the pair uv lies inside some M_i .

Let R be the set of edges of K_n lying inside no M_i ; by what we just proved, $R \subseteq I$. Counting the edges of J according to the set M_i they lie in,

$$|J| = \sum_{i=1}^p e_J(M_i) + |J \cap R| \leq \sum_{i=1}^p (a(|M_i| - 1) - 1) + |R| = \sum_{i=1}^p e_I(M_i) + |I \cap R| = |I|.$$

Here $e_J(M_i) \leq a(|M_i| - 1) - 1$ because J is sparse, and $a(|M_i| - 1) - 1 = e_I(M_i)$ because M_i is tight. $\square$

Applying Lemma 3.2 in both directions gives the following.

Corollary 3.3. *All maximal sparse subsets of $E(K_n)$ have the same size r , the largest size of a sparse subset. A graph on n vertices is τ_k -maximal if and only if it is $(k+1)$ -sparse and has exactly r edges.*

Remark 3.4. In the language of matroids, Lemma 3.2 says that the sparse sets are the independent sets of a matroid on $E(K_n)$. This is a special case of a theorem of Edmonds [1] on matroids defined by submodular functions, and such count matroids are well known in rigidity theory [3, 7]. The direct argument above keeps the note self-contained.

4. THE RANK

Taking $X = V$ in (2.1) shows $r \leq a(n-1) - 1$. For the reverse inequality we need one sparse graph with $a(n-1) - 1$ edges; Wang and Tian give a family of them [6], and the following one is quick to check.

Lemma 4.1. *Let $n \geq 2a$. Split the vertex set into A with $|A| = a$ and B with $|B| = n - a \geq a$, fix $W \subseteq B$ with $|W| = a$, and fix two vertices w, w' of W . Let G_0 have as edges all pairs inside A , all pairs between A and B , and all pairs inside W except ww' (Figure 1). Then G_0 is a -sparse and has $a(n-1) - 1$ edges.*

Proof. The number of edges is $\binom{a}{2} + a(n-a) + \binom{a}{2} - 1 = a(n-1) - 1$.

Put $C = A \cup W$. On C the graph G_0 is the complete graph on $2a$ vertices without the edge ww' , and each vertex outside C is adjacent exactly to the vertices of A . We first check (2.1) for sets $Y \subseteq C$ with $y = |Y| \geq 2$. If $y = 2a$, then $Y = C$ and $e_{G_0}(Y) = \binom{2a}{2} - 1 = a(2a-1) - 1$, which is (2.1) with equality. If $y \leq 2a-1$, then $(y-1)(2a-y) \geq 2$: for $y = 2$ the product is $2a-2 \geq 2$, and for $y \geq 3$ the first factor is at least 2 and the second at least 1. This rearranges to $y(y-1) \leq 2a(y-1) - 2$, so

$$e_{G_0}(Y) \leq \binom{y}{2} \leq a(y-1) - 1.$$

Now let $X \subseteq V$ with $|X| \geq 2$, and put $Y = X \cap C$ and $Z = X \setminus C$. There are no edges inside Z , and a vertex of Z has at most $|X \cap A| \leq \min(a, |Y|)$ neighbours in X . Hence

$$e_{G_0}(X) \leq e_{G_0}(Y) + |Z| \min(a, |Y|).$$

If $|Y| \geq 2$, this is at most $a(|Y| - 1) - 1 + a|Z| = a(|X| - 1) - 1$. If $|Y| = 1$, it is at most $|Z| \leq a|Z| - 1 = a(|X| - 1) - 1$, as $|Z| \geq 1$ and $a \geq 2$. If $Y = \emptyset$, it is $0 \leq a(|X| - 1) - 1$. $\square$

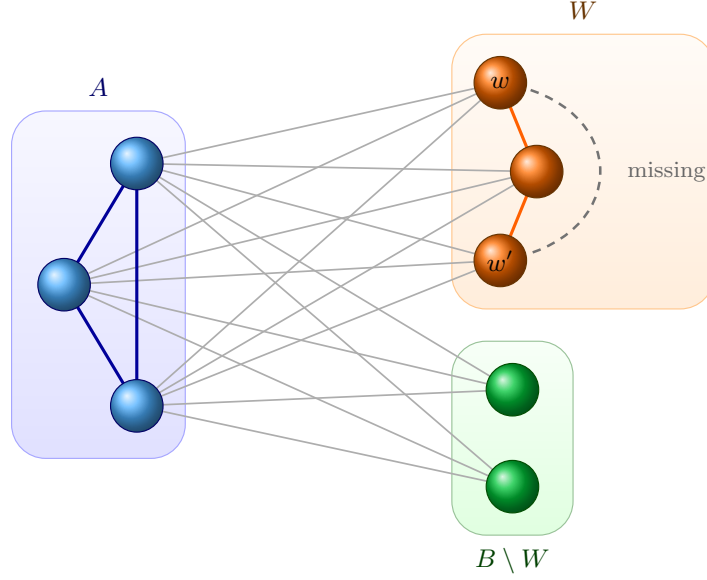


FIGURE 1. The graph G_0 for $a = 3$ and $n = 8$. On $A \cup W$ it is the complete graph K_6 without the edge ww' , and each vertex of $B \setminus W$ is joined to the three vertices of A . It has $20 = 3 \cdot 7 - 1$ edges.

Proof of Theorem 1.1. By Lemma 4.1 and the upper bound above, $r = a(n - 1) - 1 = (k + 1)(n - 1) - 1$ when $n \geq 2k + 2$. Corollary 3.3 now says that G is τ_k -maximal exactly when it is $(k + 1)$ -sparse and has $(k + 1)(n - 1) - 1$ edges, which is condition (b). $\square$

Remark 4.2. Lemma 2.1 holds for every n , and for $n \leq 2k + 1$ the whole edge set of K_n is $(k + 1)$ -sparse, since a set X with $2 \leq |X| \leq 2k + 1$ spans $\binom{|X|}{2} \leq (k + 1)(|X| - 1) - 1$ edges. So in that range the rank is $\binom{n}{2}$ and K_n is the only τ_k -maximal graph, in line with the observation of Wang and Tian. The proof above also gives back their upper bound and their result for $k = 1$.

Remark 4.3. For $k = 1$, condition (b) says $|E(G)| = 2n - 3$ and $e_G(X) \leq 2|X| - 3$ whenever $|X| \geq 2$. These are the Laman graphs, the generically minimally rigid graphs in the plane [2]. So the τ_1 -maximal graphs are exactly the Laman graphs, and in general the τ_k -maximal graphs are the bases of the $(k + 1, k + 2)$ count matroid.

Remark 4.4. Theorem 1.1 also says that a τ_k -maximal graph can be grown greedily: starting from any $(k + 1)$ -sparse graph on $n \geq 2k + 2$ vertices and adding edges one at a time as long as the graph stays sparse always ends with $(k + 1)(n - 1) - 1$ edges. Whether a given set of edges is sparse can be decided in polynomial time, for instance with the pebble game of Lee and Streinu [3].

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